

- Creating a package for an iOS app, for which a Mac machine is required
- Creating a package for an Android app, for which Android SDK and Gradle are required
- Creating a Java package for which Maven and specific JDK versions are required – Custom Jars
- Creating a hybrid app for which Ionic or Angular JS are required

We can make use of a multiple-environment setup with the help of the master and agent architecture.

The master node is where Jenkins is installed. This node performs the following tasks:

- Schedules the build jobs
- Releases the build for execution to the agent node
- Monitors various conditions such as its status (online or offline)
- Checks agents' disk space in the workspace
- Captures the build result and generates reports for it, with the master acting as the portal to view it

An agent establishes a bi-directional communication link like TCP/IP with the master in Jenkins. Thus, it distributes the work load from the master and executes jobs dispatched by it. It can run on operating systems such as Windows, Mac and Linux.

You can follow the steps given below to configure an agent node:

- Log in to the Jenkins dashboard, and click on *Manage Jenkins > Manage node*.
- To create a new agent node click on *New node*.
- Enter the name for the agent node and click on *Permanent agent*.
- Jenkins also has a provision to copy the existing node configuration once the first agent is created.
- Fill the node configuration details.
- Every agent node should have a unique name.
- **Description:** This is a short description of the agent node.
- **# of executors:** This shows the number of builds apart from the concurrent build which can be executed on this particular node.
- **Remote root directory:** Set up an

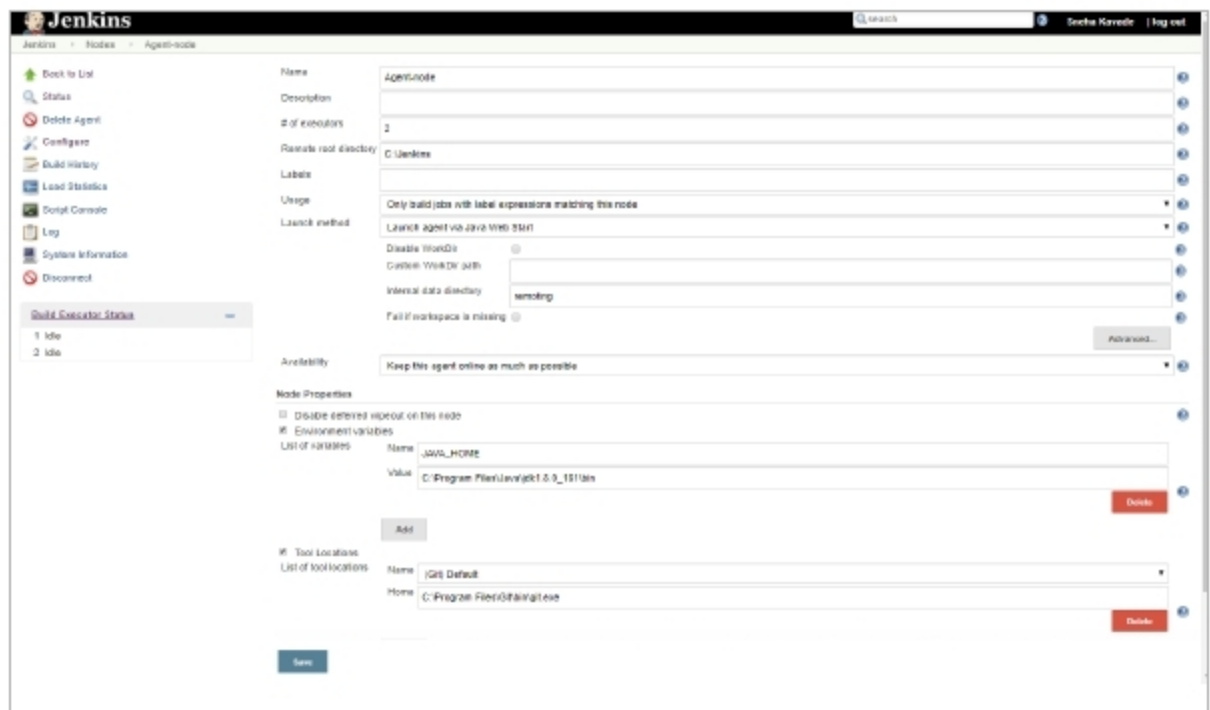


Figure 5: Node configuration

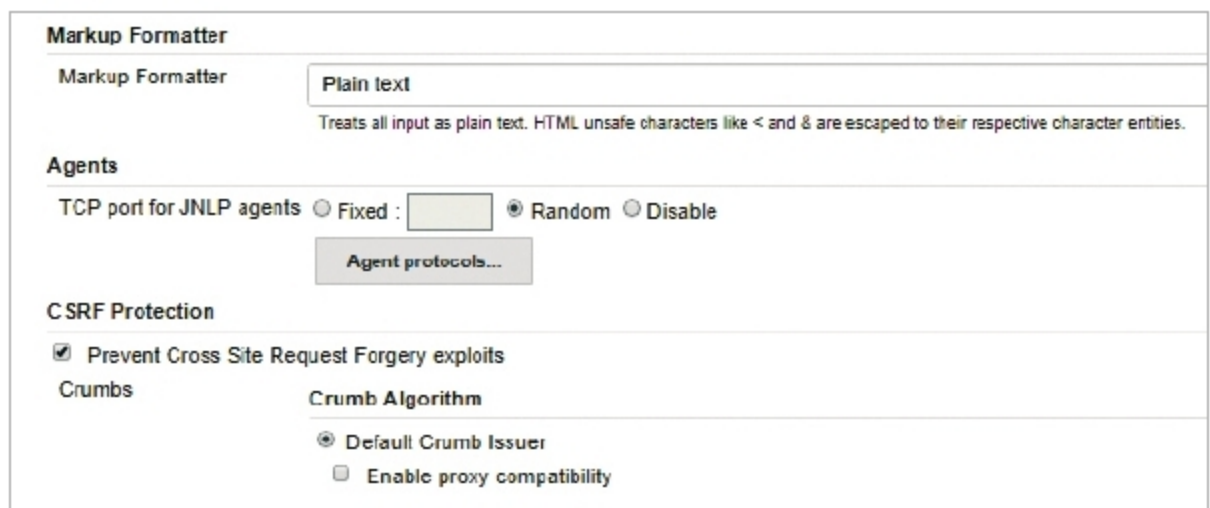


Figure 6: Agent protocol

absolute path as a working directory on the node dedicated for Jenkins.

- **Labels:** Multiple agent nodes can be logically grouped together by providing a common tag. Multiple labels must be separated by a space.
- **Usage:** Use this node as much as possible. Jenkins utilises it freely for jobs that can be executed using this node.
- **Only build jobs with label**

expressions matching this node:

Used only when the job is restricted to build on this agent.

Agents can be launched via different methods:

- Launch agent via SSH
- Launch agent via Java Web Start
- Launch agent by executing a command on the master

Here, the Java Web Start method is used to launch the agent. For that we need

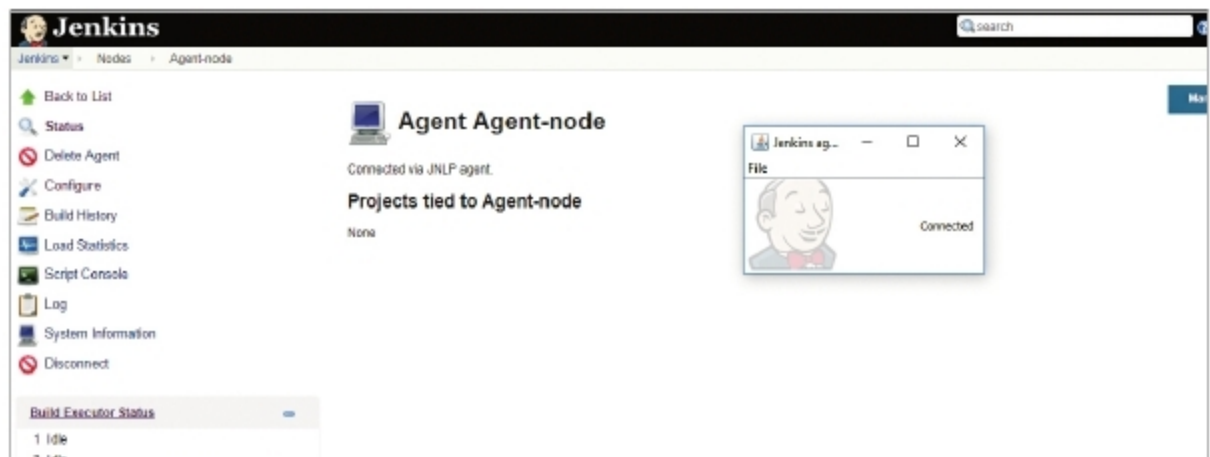


Figure 7: Agent node connected